

Konrad Szulc

Lab Work Report

Introduction to Mobile App Design and Development
R0334-3014 / Kalpana Rathod

Laurea University of Applied Sciences

13.10.2025

Lab #3: Mobile App with Ionic Framework and API call

Table of Contents

1	GitHub Repository	3
2	Learning Outcome and Reflection	3
3	Creating a movie app, with API calls	4
4	Creating my own simple app	12
	References	22
	Figures	23

1 GitHub Repository

I have created an additional 2 new repos; to guarantee there is no issues with the prior GitHub repo content submissions, this allows for grading of previous work to not be tarnished. The repo link for the Movie app is: <https://github.com/konrad-szulc/firstMovieApp>, and the repo link for my personal app is: <https://github.com/konrad-szulc/Labwork3PersonalApp>. I decided not to try to use branches, just because while I am still learning, I want to avoid as many mistakes as possible as this already is quite a bit of new material. Nonetheless, I still uploaded my report to the previous repo I provided last report, thus if needed to grab this file from there as well, the professor is free to do so.

2 Learning Outcome and Reflection

This course continues to be very interesting as there is lots of new material I have not dealt with before. Continuing to learn typescript when even my JavaScript skills are somewhat new is quite the challenge. However, it is very rewarding to see hard work pay off and noticing the connections between the functions and ion components producing a useable app. There were quite a few challenges from both tasks required, as the course continues to be lots of new information that can be hard to digest so much. I provide more detail into the specifics of each task in the respective section in this lab. Trying to understand and think of which component or creating a function that can be called upon later in typescript and ionic can be difficult to know. It is a constant test, to know what the best possible approach in a certain scenario is and when to use it. As well, constantly having to pay attention to the numerous syntax issues I would make. Luckily, VS code is great at explaining why something is a problem, and with the numerous info provided by our tutor teacher and professor for this course helps alleviate those issues. I especially appreciate understanding how a file with hooks and functions can help to be called in pages to essentially bridge the gap of getting info, but keeping it clean to understand where it all comes from rather than on a page code section itself.

As I have now formerly made groups for upcoming projects with some individuals from the course, I would continue to ask my group mates, how they understand the videos, and why we use a certain type of code. I appreciate also providing valuable tips, such as VS code having the functionality to show when redundant imports are made or are needed based on the blue blub icon. It seems others too had struggled with some of the information provided to use, thus was a great sense of camaraderie in these tasks giving motivation to keep learning and trying.

The module itself contains quite a bit of information that was best, consumed in small chunks, as it can be hard to sit through multiple 1-hour to 3-hour long videos in one go, with lots of new technical terms. I do appreciate that they tend to reiterate some new things I learned from previous modules to continue ingraining those into my mind. However, I would also get quite lost when the creators sometimes would mention that I (the audience) may already know this or seen this type of

syntax, when I have not, thus becomes my own path to seeking that information to continue learning the material.

The video by Academind (2020) goes over some key features, we have previously learned in the other modules, such as understanding how Ionic components function within the app, what exactly goes on behind the scenes to get the result when using a component such as button. The video then goes to provide some useful information by formulating a calculating BMI app while teaching along the way, such as working with grids, react hooks in typescript, differences between using html and react in Ionic like using className to add some styling to the Ionic structure. I especially appreciated how in depth the creator would go into the details of why something may not work, like how needing to use a certain bracket on the body content, since more functions were added. Furthermore, the video by Awais Mirza (2022) was similarly creating a sample app to showcase various tools a viewer may need when creating their own app. As well, a little look into native API's which utilize a device's features like the camera. All in all, while lots of material to understand, still useful for future reference

Overall, while there were some hardships, and hours' worth of making mistakes, trying to understand the process, it helps to continually hone these skills. Especially to understand how the connectivity works, better grasping the structure of Ionic Framework to then have the mindset to create a responsive, smooth app in a fraction of the time. This course has also illustrated to me, the number of improvements I can make to my coding, such as better error handling, better formatting my code for better clarity and more stylised content to be more engaging.

3 Creating a movie app, with API calls

This task was quite difficult due to the complexity of understanding so much typescript, understanding the different types, useStates, useEffects, functions, and different errors that can occur or appear. While Simon Grimm's (2023) video for navigation, components, and HTTP calls provides most of the necessary steps, and gives a nice-looking end result, following the video was quite tough due to the nature of jump cuts, and consistent errors that arise without much explanation. I really appreciate the professor for providing the API page with hooks, to completely understand how it functions, on top of some notes provided by the tutor Shuang to completely understand which even function and piece of code do. Gathering insight into good practices like using a .env for API keys, and error-handling. While this task took a lot longer than expected, due to such complex language, it definitely improved my skills at reading the structure of Ionic, and understanding, how different pages can grab information from each other, and the functionality of displaying that piece of information. I have provided screenshots with titles for better reference for later study of different aspects I had to utilize. As well, the app showed the ability to go native within chrome, acting what the app would look like on mobile. I created the needed files to simulate a mobile device; however, I do not have

an apple computer, nor android studio at the time to test the program itself. This was great experience to try to build a simple app for the next task.



Figure 1 Created .env file to house my API key for security reasons.

```

1 # See https://help.github.com/articles/ignoring-files/
2
3 # dependencies
4 /node_modules
5 /.pnp
6 .pnp.js
7
8 # testing
9 /coverage
10
11 # production
12 /dist
13
14 # misc
15 .DS_Store
16 .env.local
17 .env.development.local
18 .env.test.local
19 .env.production.local
20 /.nx
21 /.nx/cache
22 /.vscode/*
23 !/.vscode/extensions.json
24 .idea
25 .env
26
27 npm-debug.log*
28 yarn-debug.log*
29 yarn-error.log*
30
31 # Optional eslint cache
32 .eslintcache
33
```

Figure 2 Added .env to .gitignore to prevent it being pushed to the repo.

```

1 import { Redirect, Route } from 'react-router-dom';
2 import { IonApp, IonRouterOutlet, setupIonicReact } from '@ionic/react';
3 import { IonReactRouter } from '@ionic/react-router';
4 import Home from '../pages/Home';
5
6 /* Core CSS required for Ionic components to work properly */
7 import '@ionic/react/css/core.css';
8
9 /* Basic CSS for apps built with Ionic */
10 import '@ionic/react/css/normalize.css';
11 import '@ionic/react/css/structure.css';
12 import '@ionic/react/css/typography.css';
13
14 /* Optional CSS utils that can be commented out */
15 import '@ionic/react/css/padding.css';
16 import '@ionic/react/css/float-elements.css';
17 import '@ionic/react/css/text-alignment.css';
18 import '@ionic/react/css/text-transformation.css';
19 import '@ionic/react/css/flex-utils.css';
20 import '@ionic/react/css/display.css';
21
22 /**
23  * Ionic Dark Mode
24  * ---
25  * For more info, please see:
26  * https://ionicframework.com/docs/theming/dark-mode
27  */
28
29 /* Import '@ionic/react/css/palettes/dark.always.css'; */
30 /* Import '@ionic/react/css/palettes/dark.class.css'; */
31 import '@ionic/react/css/palettes/dark.system.css';
32
33 /* Theme variables */
34 import '../theme/variables.css';
35 import Details from '../pages/Details';
36
37 setupIonicReact();
38
39 const App: React.FC = () => {
40   <IonApp>
41     <IonReactRouter>
42       <Route exact path="/movies" component={Home}></Route>
43       <Route exact path="/movies/:id" component={Details}></Route>
44       <Route exact path="/" />
45       <Redirect to="/movies" />
46     </IonReactRouter>
47   </IonApp>
48 };
49
50 export default App;
51

```

Figure 3 Routing in App page, to create navigation between the pages.

```

1 // useApi hook to use api
2 // Follow this video for details https://youtu.be/zn-apt72n30?list=PL486
3 // Custom hook to interact with the IMDb API
4
5 // TypeScript custom union for search types (optional)
6 export enum SearchType {
7   all = "",
8   movie = "movie",
9   series = "series",
10  episode = "episode",
11 }
12
13 // Interface for search result items
14 export interface SearchResult {
15   title: string;
16   year: string;
17   poster: string;
18   imdbId: string;
19   type: string;
20 }
21
22 // Interface for search error response
23 export interface SearchError {
24   response: string;
25   error: string;
26 }
27
28 // Interface for detailed movie information
29 export interface DetailsResult {
30   genre: string;
31   title: string;
32   year: string;
33   poster: string;
34   plot: string;
35   awards: string;
36   imdbRating: string;
37   direction: string;
38   actors: string;
39   website: string;
40 }
41
42 // Custom hook to interact with the IMDb API
43 export const useImdb = () => {
44   const url = "https://www.omdbapi.com/";
45   const apiKey = import.meta.env.VITE_MOVIE_API_KEY; // following good practice by using .env file ignored by git to not publish my api key public
46   // Please use the env file of .env with your own API key as MOVIE_API_KEY = "your API key here" to use the useImdb function, do not want to showcase my personal API
47   // function to search for movies, series, or episodes by title and type
48   const searchData = async (title: string, type: SearchType): Promise<SearchResult[] | SearchError> => {
49     const result = await fetch(
50       `${url}?s=${encodeURIComponent(title)}&type=${type}&apikey=${apiKey}`
51     );
52     return result.json();
53   };
54
55   // function to get detailed information about a specific movie, series, or episode by IMDb ID
56   const getDetails = async (id: string): Promise<DetailsResult> => {
57     const result = await fetch(`${url}?i=${id}&plot=full&apikey=${apiKey}`);
58     return result.json();
59   };
60
61   // return the functions to be used in components
62   return {
63     searchData,
64     getDetails,
65   };
66 };
67
68 export default useImdb;

```

Figure 4 Created useApi with hooks page to grab the information from the API to use on pages.

```

1 import { IonAvatar, IonContent, IonHeader, IonIcon, IonItem, IonList, IonLabel, IonList, IonPage, IonSearchbar, IonSelect, IonSelectOption, IonTitle, IonToolbar, useIonAlert, useIonLoading } from '@ionic/react';
2 import { useIonRouter } from '@ionic/react-router';
3 import { useIonRouter } from '@ionic/react-router';
4 import { useIonRouter } from '@ionic/react-router';
5
6 const Home: React.FC = () => {
7
8   const [searchData, setSearchData] = useState<SearchData>({});
9
10  const [searchForm, setSearchForm] = useState<SearchForm>({});
11  const [type, setType] = useState<SearchType>('all');
12  const [results, setResults] = useState<SearchResult[]>([]);
13  const [loading, setLoading] = useState<boolean>();
14  const [presentAlert, setPresentAlert] = useState<boolean>();
15
16  useEffect(() => {
17    if (searchForm === '') {
18      setResults([]);
19    }
20  }, [searchForm]);
21
22  const loadData = async () => {
23    await IonLoading({
24      message: 'Loading...',
25      duration: 1000,
26      cssClass: 'my-custom-class',
27    });
28    const result = await searchData(searchForm, type);
29    console.log('the result is:', result);
30    await IonLoading({
31      message: 'Loading...',
32      duration: 1000,
33      cssClass: 'my-custom-class',
34    });
35    if (result?.error) {
36      setPresentAlert(true);
37    } else {
38      setResults(result.results);
39    }
40  };
41
42  loadData();
43  return (
44    <IonPage>
45      <IonHeader>
46        <IonToolbar color="primary">
47          <IonTitle>Movie App</IonTitle>
48        </IonToolbar>
49      </IonHeader>
50      <IonContent>
51        <IonSearchbar
52          value={searchForm}
53          debounce={1000}
54          onInputChange={(e) => setSearchForm(e.detail.value)}
55        </IonSearchbar>
56        <IonList>
57          <IonLabel>Search Type</IonLabel>
58          <IonSelect
59            value={type}
60            onIonChange={(e) => setType(e.detail.value)} slot="end">
61            <IonSelectOption value="all">All</IonSelectOption>
62            <IonSelectOption value="movie">Movie</IonSelectOption>
63            <IonSelectOption value="series">Series</IonSelectOption>
64            <IonSelectOption value="episode">Episode</IonSelectOption>
65          </IonSelect>
66        </IonList>
67        <IonList>
68          {results.map((item: SearchResult) => (
69            <IonItem>
70              <IonAvatar slot="start">
71                <IonImg src={item.poster}></IonImg>
72              </IonAvatar>
73              <IonLabel className="ion-text-wrap">{item.title}</IonLabel>
74
75              {item.type === 'movie' && <IonIcon slot="end" icon={videocamOutline}/>}
76              {item.type === 'series' && <IonIcon slot="end" icon={desktopOutline}/>}
77              {item.type === 'game' && <IonIcon slot="end" icon={gameControllerOutline}/>}
78            </IonItem>
79          ))}
80        </IonList>
81      </IonContent>
82    </IonPage>
83  );
84
85  export default Home;
86
87

```

Figure 5 Home page, structure composed to display the information.

```

61 <IonSelectOption value="episode">Episode</IonSelectOption>
62 </IonSelect>
63
64 </IonItem>
65
66 <IonList>
67   {results.map((item: SearchResult) => (
68     <IonItem button key={item.imdbID} routerLink={`/movies/${item.imdbID}`}>
69       <IonAvatar slot="start">
70         <IonImg src={item.poster}></IonImg>
71       </IonAvatar>
72       <IonLabel className="ion-text-wrap">{item.title}</IonLabel>
73
74       {item.type === 'movie' && <IonIcon slot="end" icon={videocamOutline}/>}
75       {item.type === 'series' && <IonIcon slot="end" icon={desktopOutline}/>}
76       {item.type === 'game' && <IonIcon slot="end" icon={gameControllerOutline}/>}
77     </IonItem>
78   ))}
79 </IonList>
80 </IonContent>
81 </IonPage>
82
83 ];
84
85 export default Home;
86
87

```

Figure 6 Home page part 2

```

1 import { IonBackButton, IonButton, IonButtons, IonCard, IonCardContent, IonCardHeader, IonCardSubtitle, IonCardTitle, IonContent, IonFooter, IonHeader, IonIcon, IonImg, IonItem, IonLabel, IonModal, IonPage, IonTitle, IonToolbar, IonViewWrapper } from '@ionic/react';
2 import { useState } from 'react';
3 import { RouterComponentProps } from 'react-router';
4 import { useHistory } from 'react-router-dom';
5 import { IonBackButton, IonHeader, IonPage, IonTitle, IonToolbar, IonViewWrapper } from '@ionic/react';
6
7 interface DetailsPageProps extends RouterComponentProps { id: string; }
8
9 const Details = ({ match }: DetailsPageProps) => {
10   const [getDetails] = useState();
11   const [information, setInformation] = useState<DetailsResult | undefined>(undefined);
12
13   useViewWillEnter(() => {
14     const fetchDetails = async () => {
15       const id = match.params.id;
16       const data = await getDetails(id);
17       setInformation(data);
18       console.log("data: ", data);
19     }
20     fetchDetails();
21   })
22
23   return (
24     <IonPage>
25       <IonHeader>
26         <IonButtons slot="start">
27           <IonBackButton defaultHref="/movies"/></IonBackButton>
28         </IonButtons>
29         <IonTitle>{information?.name}</IonTitle>
30       </IonHeader>
31       <IonContent>
32         <Information AS (
33           <IonCard>
34             <IonCardHeader>
35               <IonCardTitle>{information?.title}</IonCardTitle>
36               <IonCardSubtitle>{information?.year}</IonCardSubtitle>
37             </IonCardHeader>
38             <IonCardContent>
39               <IonImg src={information?.poster}</IonImg>
40               <IonText>{information?.name}</IonText>
41               <IonText>{information?.plot}</IonText>
42               <IonText>{information?.director}</IonText>
43             </IonCardContent>
44           </IonCard>
45         )
46         <IonModal trigger="open-modal" initialBreakpoints={0.2} breakpoints={0.2, 0.5, 0.75}>
47           <IonContent className="ion-padding">
48             <IonList>
49               <IonListHeader>
50                 <IonText>{information?.name}</IonText>
51               </IonListHeader>
52               <IonList>
53                 <IonText>{information?.plot}</IonText>
54                 <IonText>{information?.director}</IonText>
55                 <IonText>{information?.year}</IonText>
56                 <IonText>{information?.actors}</IonText>
57               </IonList>
58             </IonContent>
59           </IonModal>
60         <IonFooter>
61           <IonButton expand="full" id="open-modal">Show More</IonButton>
62         </IonFooter>
63       </IonPage>
64     )
65   );
66 }
67
68 export default Details;

```

Figure 7 Details page, to then dive deeper into a search, providing more information.

```

1 <IonListHeader>
2   <IonText>{information?.name}</IonText>
3 </IonListHeader>
4 <IonList>
5   <IonText>{information?.plot}</IonText>
6   <IonText>{information?.director}</IonText>
7   <IonText>{information?.year}</IonText>
8   <IonText>{information?.actors}</IonText>
9 </IonList>
10 </IonContent>
11 </IonModal>
12 </IonContent>
13 </IonFooter>
14 <IonButton expand="full" id="open-modal">Show More</IonButton>
15 </IonFooter>
16 </IonPage>
17 )
18 }
19
20 export default Details;

```

Figure 8 Details part 2

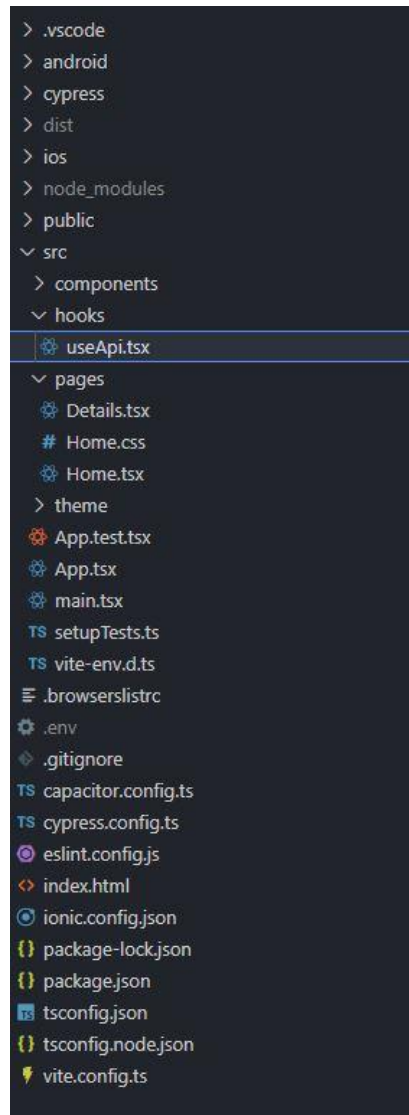


Figure 9 Final amount of files needed and created.

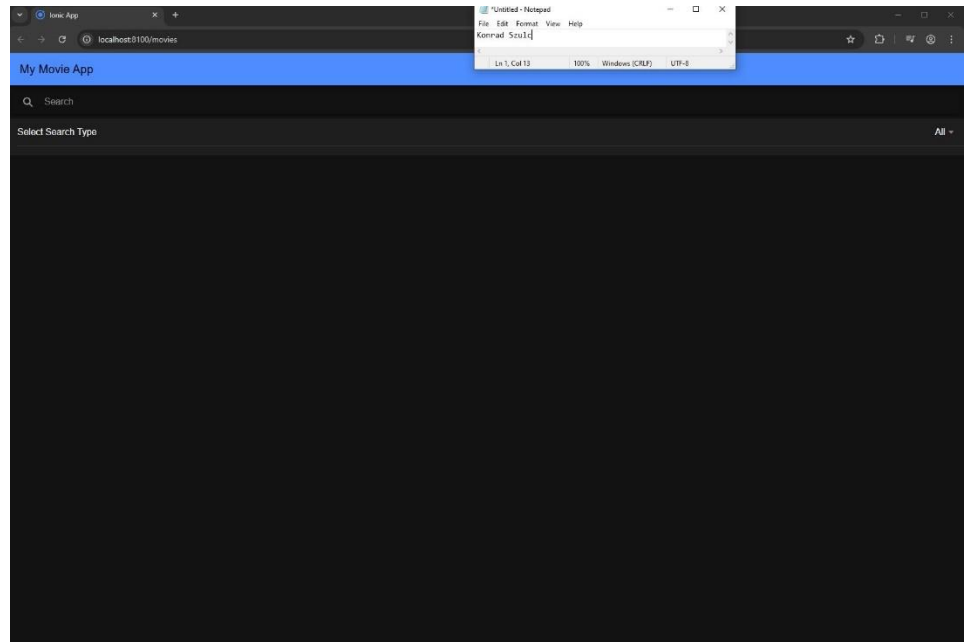


Figure 10 Example of home page unsearched.

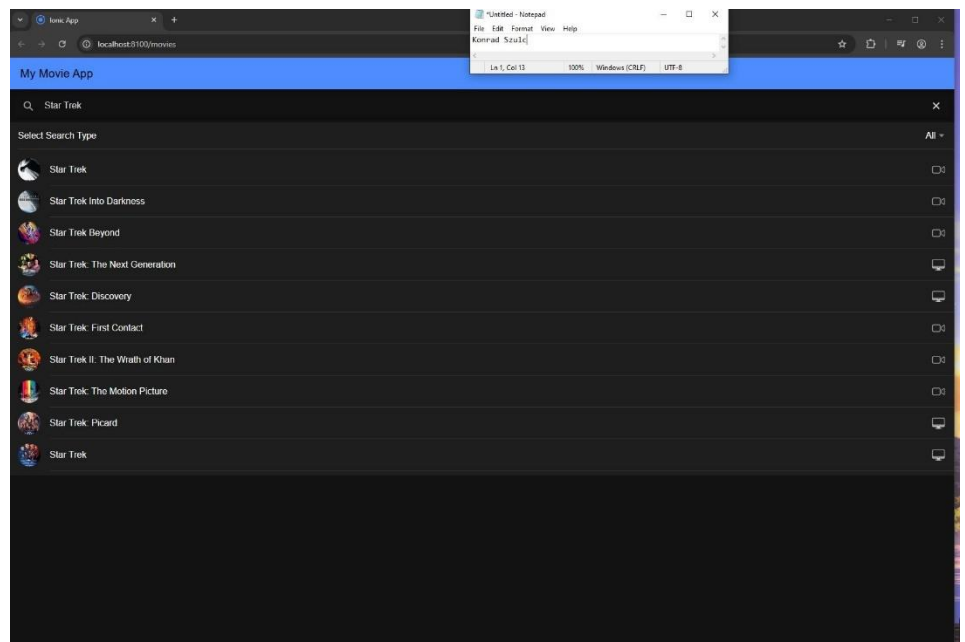


Figure 11 Example of home page searched with star trek.



Figure 12 Details when clicked on search.

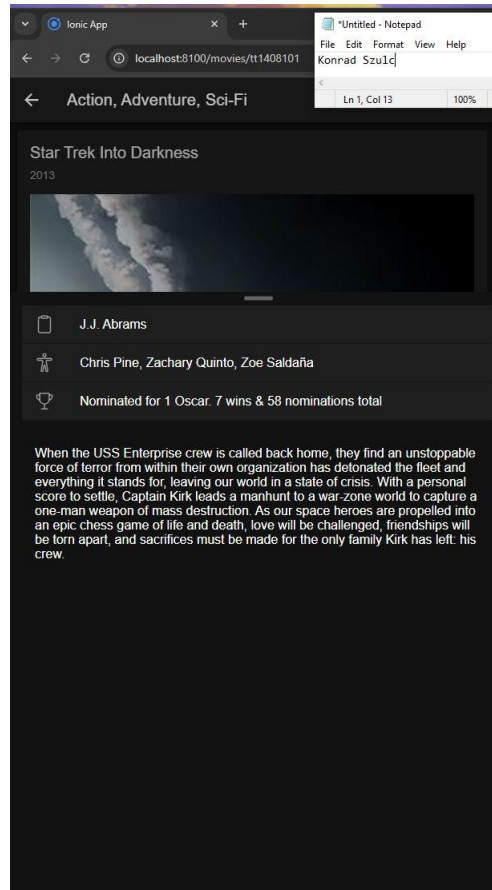


Figure 13 Modal introduced on details page to bring up even more information provided by API.

4 Creating my own simple app

First, I started by brainstorming the simple layout of my app, and what exactly was the purpose. Originally the plan was to create a simple app with a feature to add user text as a note or list for my favourite games. It would include a simple welcome page with a button to dive into 3 tabs, the 3 tabs were About, My Game List, and Settings. About would have very brief description as to what the app is about, with the Game list having the main functionality of adding a note. Lastly, the settings tab was to have any settings such as dark mode or enable notifications. As seen in figure 14 I created exactly these pages with functional code for menu navigation across the whole app for smooth user experience. However, I decided I wanted to pivot and try to use the API call as we just learned from the first task and apply it to another API.

I decided using something interesting that I can enjoy myself whenever I want, thus I registered for a free key from <https://thecatapi.com/>. This allows me to gather a simple random cat picture to then display on the app. I would create a button that would call this API from the hook folder and display a random cat pic. Which also

meant for clarity I then had to change the file names, tab name, imports, on each page just for understanding when looking at the app in the future. A few notable things I learned while trying to create this app from the previous knowledge, and the new knowledge acquired through the module. Firstly, the `IonMenu` had to be created in the `App.tsx` file for a smooth experience to be persistent throughout the app. I tried creating it in each page, but it would be laggy, and not function correctly, mainly cause the `<IonApp>` should only be used in one page. Next, I used the same tip from the first app by our tutor, it is good practice to use a `.env` file that is not saved to the GitHub public repo. Thus, must use your own API substituted into the application to run properly in an env file. As well, while there were some similarities in the API file with needed hooks and process of showing the image similar to the movie app, there were some requirements the API required to function, such as being called by headers, luckily the API provided some examples shown in figure 15 and even though they were with different language. Some trial and error with the previous code, understanding what exactly `useState`, and different type script functions got it too work.

However, a stark difference I noticed compared to the previous API call app, is the difficulty of error handling, and loading problems. While the app runs well with the API working, I recognize the importance of making sure there is error handling in case the program does not fetch the accurate information, while the `thecatapi` website shows some error handling, as well previously our tutor Shuang provides some examples of error handling, I had difficulties understanding where to include to avoid messing up the actual result. I recognize I am still learning and developing an app takes time. Furthermore, the same can be said about creating a dark mode, for while Ionic provides a doc about utilizing dark mode, I have not had much experience with the CSS portion of the Ionic app and require more hands-on experience and more to learn. As seen in the Figure 25 page I got to toggle to function.

Lastly, a very useful resource when trying to have different icons, or even just different UI components are the Ionic docs. Specifically in this case were <https://ionicframework.com/docs/components> and <https://ionic.io/ionicons>. This provided some more personal touch to my app, as well to develop different components like a toggle or menu. Ultimately, though it was a new experience, requiring lots of patience and testing, I am happy to continue learning to develop even better apps.

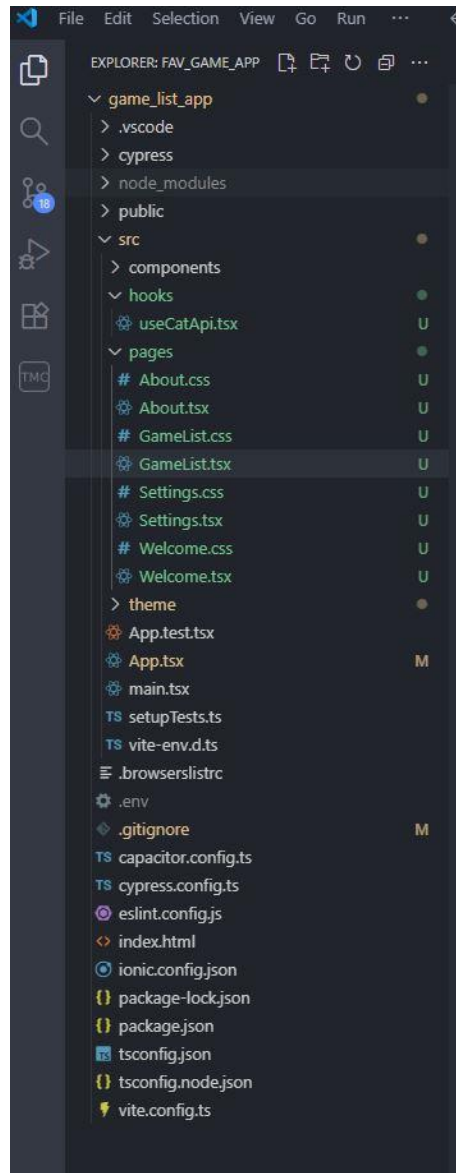


Figure 14 Original names when making GameList app instead, changed to CatPic.

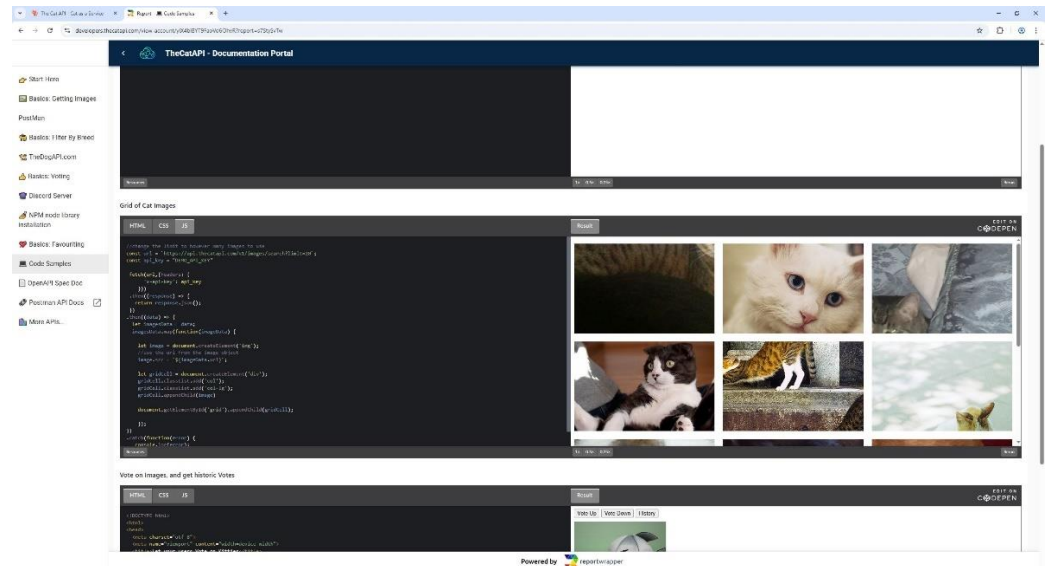


Figure 15 thecatapi website providing some JavaScript code examples for grabbing API info.

```

1 import { useState } from 'react';
2
3
4 export const useCatApi = () => {
5
6   const url = 'https://api.thecatapi.com/v1/images/search'
7
8   //pulls apikey from .env to make sure safety of apikey not in public repo PLEASE INSERT OWN API KEY HERE TO FUNCTION
9   //can grab api key from https://thecatapi.com/
10
11   const apiKey = import.meta.env.VITE_cat_Api_Key
12
13   //creating state of catImage, which is declared as null originally and will be string
14   const [catImage, setCatImage] = useState<string | null>(null);
15   //function to get a random cat image from the api
16   const fetchCatImage = async () => {
17     //fetch request for api data, waits for server to respond before attributing it to response variable, required to do through headers
18     const response = await fetch(url, {
19       headers: {
20         'x-api-key': apiKey
21       }
22     });
23     //waits for the fetch of response to be through the json function and attributed to data
24     const data = await response.json();
25     //changes the state of catImage to the correct url of the image pulled from the api in data variable
26     setCatImage(data[0].url);
27   };
28
29   return {
30     //this allows to then call these functions and variables in the catapi page
31     catImage,
32     fetchCatImage,
33   };
34 };
35
36 export default useCatApi

```

Figure 16 My useCatApi with hooks page to be used by the pages in the app.



Figure 17 Added .env with API key and added .env to .gitignore.

```
<IonMenuToggle>
  <IonItem lines="none">
    <IonButton routerLink = "/Welcome">Click here to Leave App</IonButton>
  </IonItem>
</IonMenuToggle>

</IonList>
</IonContent>
</IonMenu>

<IonTabs>
  <IonRouterOutlet id = "main-content">
    <Route exact path="/About">
      <About />
    </Route>
    <Route exact path="/CatPic">
      <CatPic />
    </Route>
    <Route path="/Settings">
      <Settings />
    </Route>
    <Route path = "/Welcome">
      <Welcome />
    </Route>
    <Route exact path="/">
      <Redirect to="/Welcome" />
    </Route>
  </IonRouterOutlet>

  <IonTabBar slot="bottom">
    <IonTabButton tab="About" href="/About">
      <IonIcon aria-hidden="true" icon={newspaperOutline} />
      <IonLabel>About</IonLabel>
    </IonTabButton>
    <IonTabButton tab="CatPic" href="/CatPic">
      <IonIcon aria-hidden="true" icon={imageOutline} />
      <IonLabel>Generate Cat Picture</IonLabel>
    </IonTabButton>
    <IonTabButton tab="Settings" href="/Settings">
      <IonIcon aria-hidden="true" icon={cogOutline} />
      <IonLabel>Settings</IonLabel>
    </IonTabButton>
  </IonTabBar>
</IonTabs>
</IonReactRouter>
</IonApp>
  </>
};
//Using the menu within the main app page for it to be persistent through out all tabs other
export default App;
```

Figure 18 Routing information properly organized to make sure no issues with navigation. Included tabs in app page due to selecting tabs template.

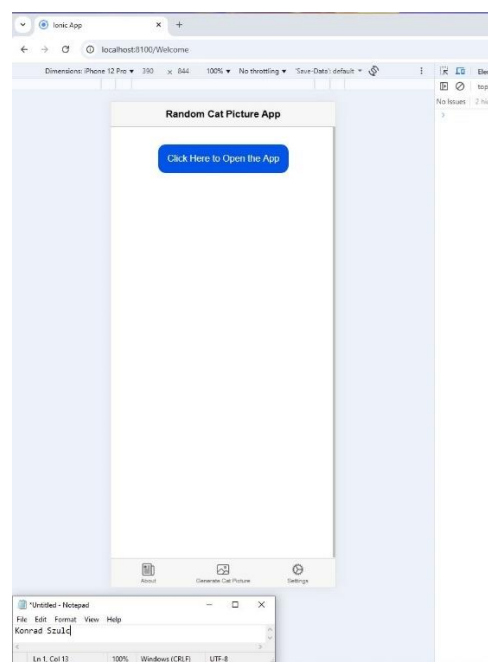


Figure 19 Example of welcome page.


```
1 import { IonButton, IonCol, IonContent, IonGrid, IonHeader, IonPage, IonRow, IonTitle, IonToolbar } from '@ionic/react';
2 import React from 'react';
3
4 const Welcome: React.FC = () => {
5
6   return (
7     <IonPage>
8       <IonHeader>
9         <IonToolbar>
10           <IonTitle className = "ion-text-center">Random Cat Picture App</IonTitle>
11         </IonToolbar>
12       </IonHeader>
13       <IonContent className="ion-padding">
14         <IonGrid>
15           <IonRow>
16             <IonCol className = "ion-text-center">
17               <IonButton routerLink = "/About">
18                 Click Here to Open the App
19               </IonButton>
20             </IonCol>
21           </IonRow>
22         </IonGrid>
23       </IonContent>
24     </IonPage>
25   );
26 };
27
28 export default Welcome;
```

Figure 20 Welcome page code.

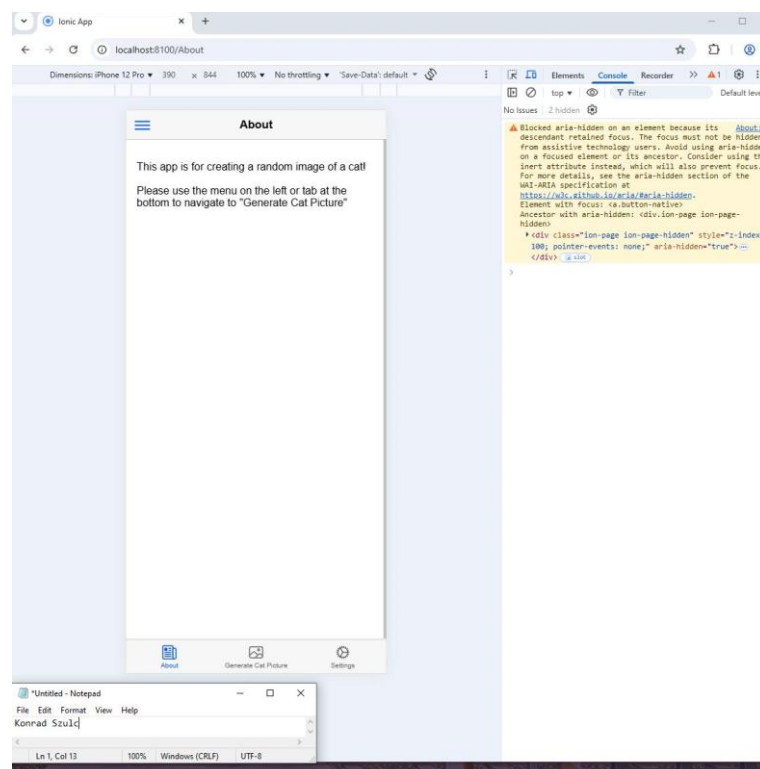


Figure 21 Example of About page.

```
1 import { IonButtons, IonContent, IonHeader, IonMenuButton, IonPage, IonTitle, IonToolbar } from '@ionic/react';
2 import './About.css';
3
4
5 const About: React.FC = () => {
6   return (
7     <IonPage>
8       <IonHeader>
9         <IonToolbar>
10           <IonButtons slot="start">
11             <IonMenuButton></IonMenuButton>
12           </IonButtons>
13           <IonTitle>About</IonTitle>
14         </IonToolbar>
15       </IonHeader>
16       <IonContent>
17         <p className="ion-padding">
18           This app is for creating a random image of a cat! <br/><br/>
19           Please use the menu on the left or tab at the bottom to navigate to "Generate Cat Picture"
20         </p>
21       </IonContent>
22     </IonPage>
23   );
24 };
25
26 export default About;
27
```

Figure 22 About page code.

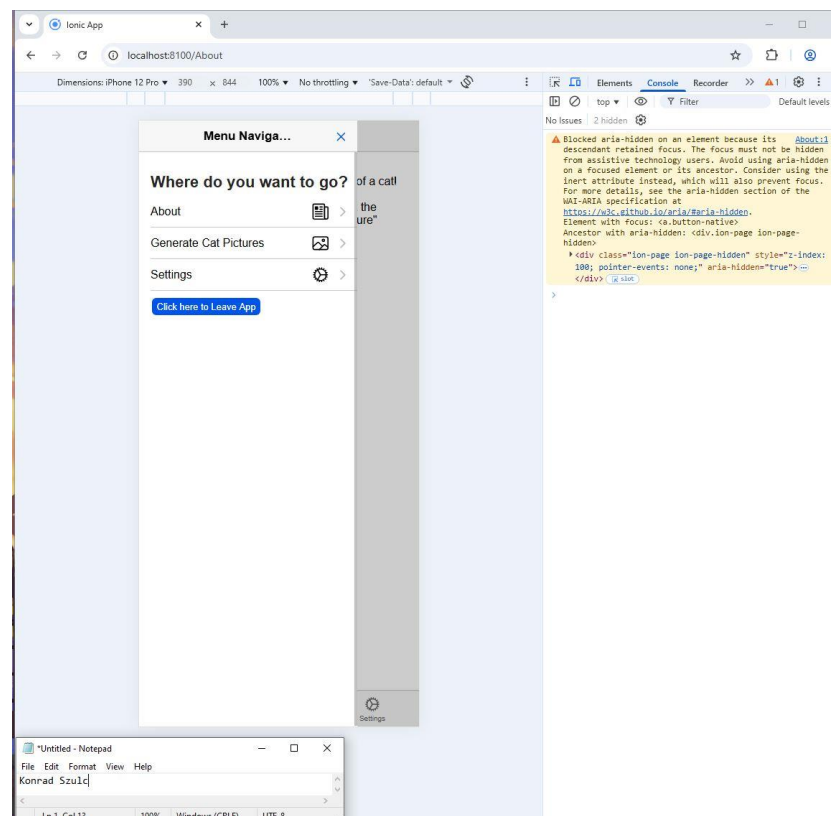


Figure 23 Example of Menu across the application.

```
const App: React.FC = () => {  
  return(  
    <IonReactRouter>  
      <IonMenu contentId="main-content">  
        <IonHeader>  
          <IonToolbar>  
            <IonTitle>Menu Navigation</IonTitle>  
            <IonButtons slot="end">  
              <IonMenuToggle autoHide>  
                <Icon icon={closeOutline}></Icon>  
              </IonButtons>  
            </IonToolbar>  
          </IonHeader>  
          <IonContent>  
            <IonList>  
              <IonListHeader>  
                <IonLabel>Where do you want to go?</IonLabel>  
              </IonListHeader>  
              <IonMenuToggle autoHide>  
                <IonItem button routerLink="/About">  
                  <Icon slot="end" icon={newspaperOutline}></Icon>  
                  <IonLabel>About</IonLabel>  
                </IonItem>  
                <IonMenuToggle>  
                  <IonItem button routerLink="/CatPic">  
                    <Icon slot="end" icon={imageOutline}></Icon>  
                    <IonLabel>Generate Cat Pictures</IonLabel>  
                  </IonItem>  
                  <IonMenuToggle>  
                    <IonItem button routerLink="/Settings">  
                      <Icon slot="end" icon={cogOutline}></Icon>  
                      <IonLabel>Settings</IonLabel>  
                    </IonItem>  
                  </IonMenuToggle>  
                </IonMenuToggle>  
              </IonList>  
            </IonContent>  
          </IonMenu>  
        </IonReactRouter>  
      </App>  
    )  
  )  
}
```

Figure 24 Menu in App page to keep persistent in the application.

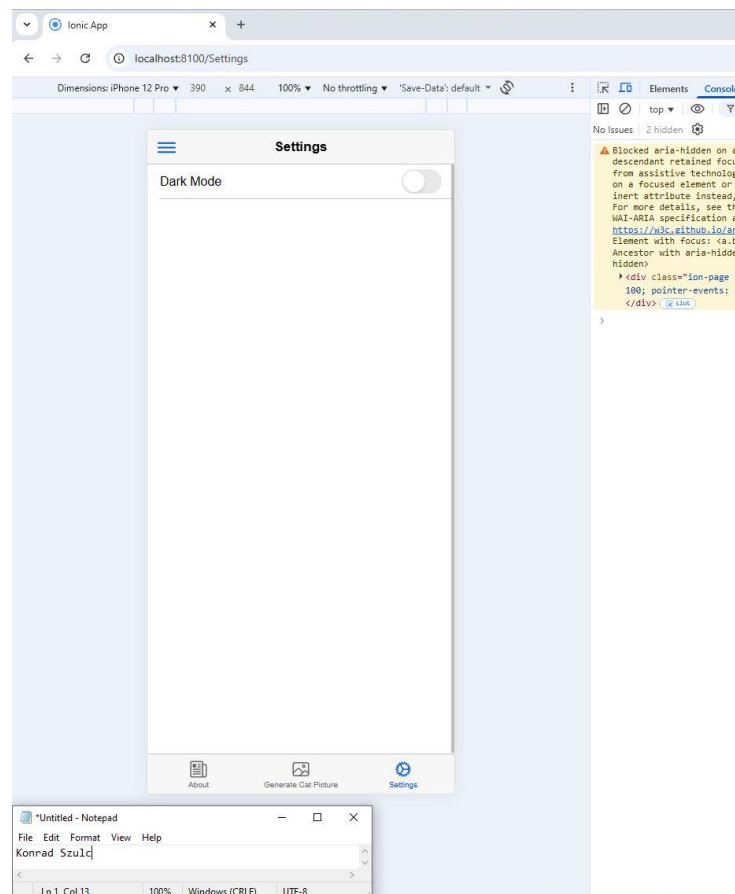


Figure 25 Setting page, includes toggle, but does not change to dark mode.

```
1 import { IonButtons, IonContent, IonHeader, IonItem, IonLabel, IonList, IonMenuButton, IonPage, IonTitle, IonToggle, IonToolbar } from '@ionic/react';
2 import './Settings.css';
3
4 const Settings: React.FC = () => {
5   return (
6     <IonPage>
7       <IonHeader>
8         <IonToolbar>
9           <IonButtons slot="start">
10             <IonMenuButton/></IonMenuButton>
11           </IonButtons>
12           <IonTitle>Settings</IonTitle>
13         </IonToolbar>
14       </IonHeader>
15       <IonContent>
16         <IonList>
17           <IonItem>
18             <IonToggle slot="end"></IonToggle>
19             <IonLabel>Dark Mode</IonLabel>
20           </IonItem>
21         </IonList>
22       </IonContent>
23     </IonPage>
24   );
25 };
26
27 export default Settings;
28
```

Figure 26 Settings page code.

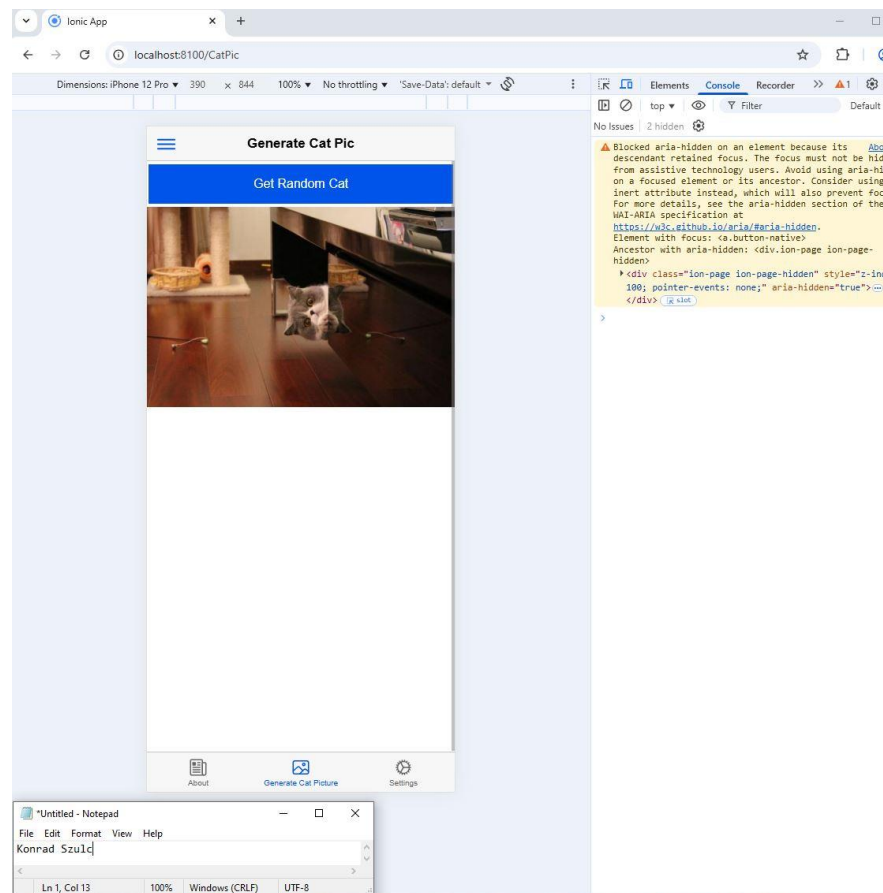


Figure 27 Example of Cat pic generator working as intended.

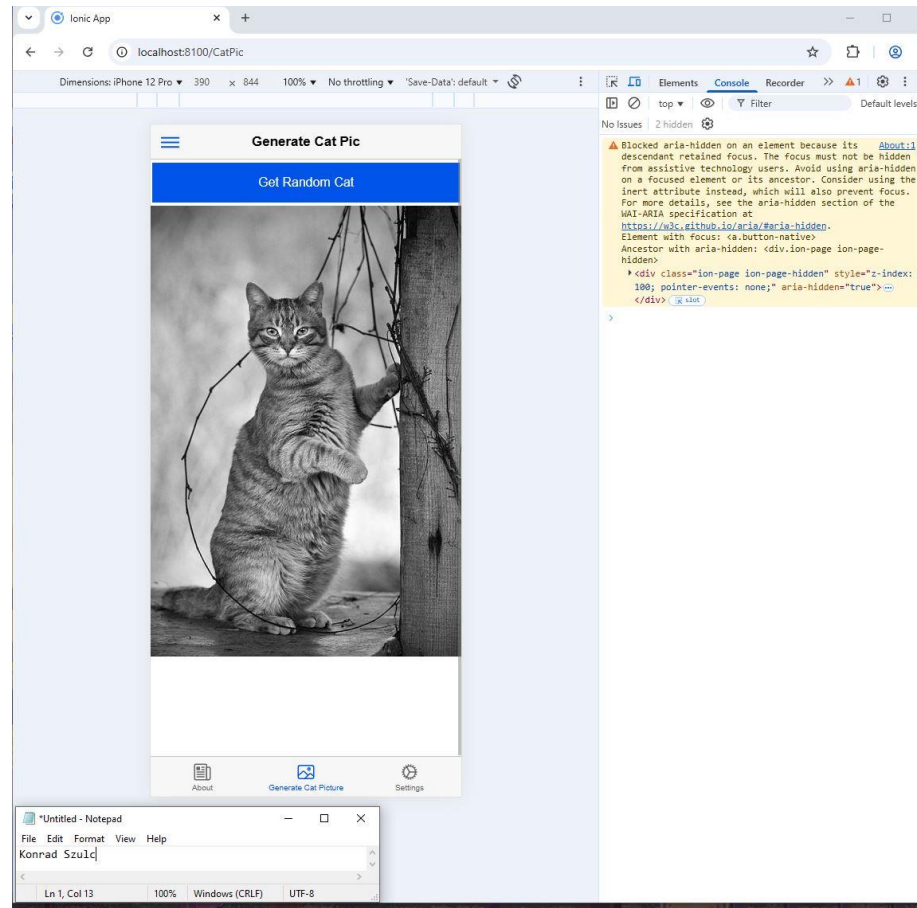


Figure 28 Another Example of using button multiple times.

```

1 import { IonButton, IonButtons, IonContent, IonHeader, IonImg, IonMenuButton, IonPage, IonTitle, IonToolbar } from '@ionic/react';
2 import './CatPic.css';
3 import { useCatApi } from '../hooks/useCatApi';
4
5 const CatPic: React.FC = () => {
6   const { catImage, fetchCatImage } = useCatApi();
7   return (
8     <IonPage>
9       <IonHeader>
10        <IonToolbar>
11          <IonButtons slot="start">
12            <IonMenuButton></IonMenuButton>
13          </IonButtons>
14          <IonTitle>Generate Cat Pic</IonTitle>
15        </IonToolbar>
16      </IonHeader>
17      <IonContent>
18        <IonButton expand="full" onClick={fetchCatImage}>
19          Get Random Cat
20        </IonButton>
21        {catImage && prevents IonImg from loading if no value in catImage, has to be string to load?}
22        {catImage && (
23          <IonImg src={catImage} alt="Random Cat Picture"></IonImg>
24        )}
25      </IonContent>
26    </IonPage>
27  );
28 };
29
30 export default CatPic;
31

```

Figure 29 Cat Pic Page code.

References

Academind. 2020. Ionic + react - tutorial for beginners. YouTube. Accessed on October 13, 2025. https://www.youtube.com/watch?v=_03VKmdrxV8

Awais Mirza. 2022. Ionic framework for react developers - how to build mobile application with react. YouTube. Accessed on October 13, 2025. <https://www.youtube.com/watch?v=O0hu752Teig>

Grimm, S. 2023. Ionic react beginners guide | navigation, components & http calls. YouTube. Accessed October 13, 2025. <https://www.youtube.com/watch?v=xn-qpnT2n3Q>

Figures

Figure 1 Created .env file to house my API key for security reasons.	5
Figure 2 Added .env to .gitignore to prevent it being pushed to the repo.	5
Figure 3 Routing in App page, to create navigation between the pages.	6
Figure 4 Created useApi with hooks page to grab the information from the API to use on pages.	6
Figure 5 Home page, structure composed to display the information.	7
Figure 6 Home page part 2.....	7
Figure 7 Details page, to then dive deeper into a search, providing more information.	8
Figure 8 Details part 2	8
Figure 9 Final amount of files needed and created.	9
Figure 10 Example of home page unsearched.	10
Figure 11 Example of home page searched with star trek.	10
Figure 12 Details when clicked on search.	11
Figure 13 Modal introduced on details page to bring up even more information provided by API. ...	12
Figure 14 Original names when making GameList app instead, changed to CatPic.	14
Figure 15 thecatapi website providing some JavaScript code examples for grabbing API info.	15
Figure 16 My useCatApi with hooks page to be used by the pages in the app.	15
Figure 17 Added .env with API key and added .env to .gitignore.	15
Figure 18 Routing information properly organized to make sure no issues with navigation. Included tabs in app page due to selecting tabs template.	16
Figure 19 Example of welcome page.	16
Figure 20 Welcome page code.	17
Figure 21 Example of About page.	17
Figure 22 About page code.	18
Figure 23 Example of Menu across the application.	18
Figure 24 Menu in App page to keep persistent in the application.	19
Figure 25 Setting page, includes toggle, but does not change to dark mode.	19
Figure 26 Settings page code.	20
Figure 27 Example of Cat pic generator working as intended.	20
Figure 28 Another Example of using button multiple times.	21
Figure 29 Cat Pic Page code.	21